Лабораторная работа 5

Построение графиков

Цвелев С.А. НПИбд-01-22

Цель работы

Основная цель работы — освоить синтаксис языка Julia для построения графиков.

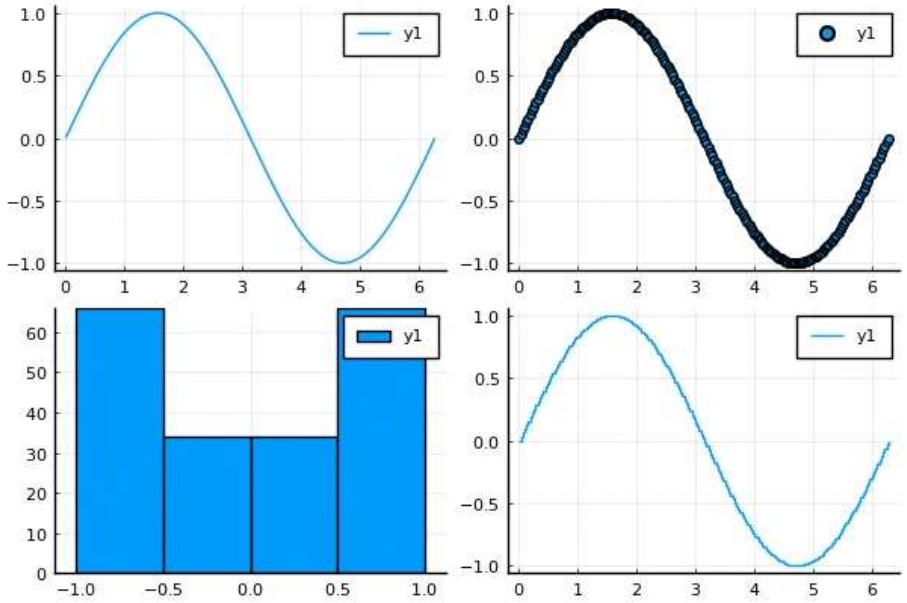
Различные типы графиков

```
[18]: x = range(0, 2pi, length=200)
      y = sin.(x)

      p1 = plot(x, y)
      p2 = scatter(x, y)
      p3 = histogram(y)
      p4 = plot(x, y, st=:step)

      plot(p1, p2, p3, p4, layout=(2,2))
```

[18]:



Грифик функции y = sin(x), x = 0, 2π

```
[20]: x = 0:0.01:2pi
      y = sin.(x)

      plot(x,y,ls=:solid)
      plot!(x,y,ls=:dash)
      plot!(x,y,ls=:dot)
      plot!(x,y,ls=:dashdot)
```

[20]:

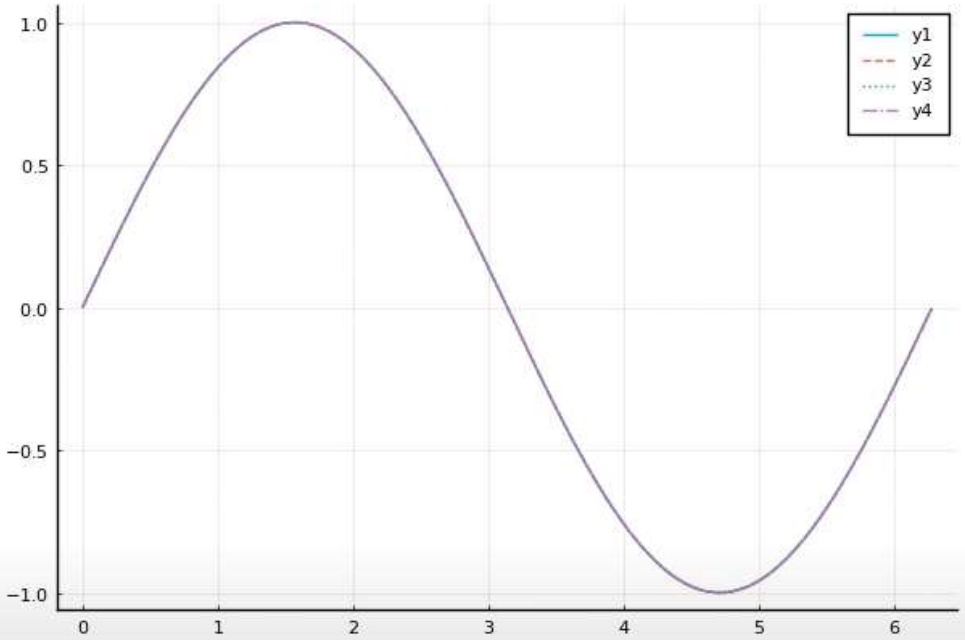
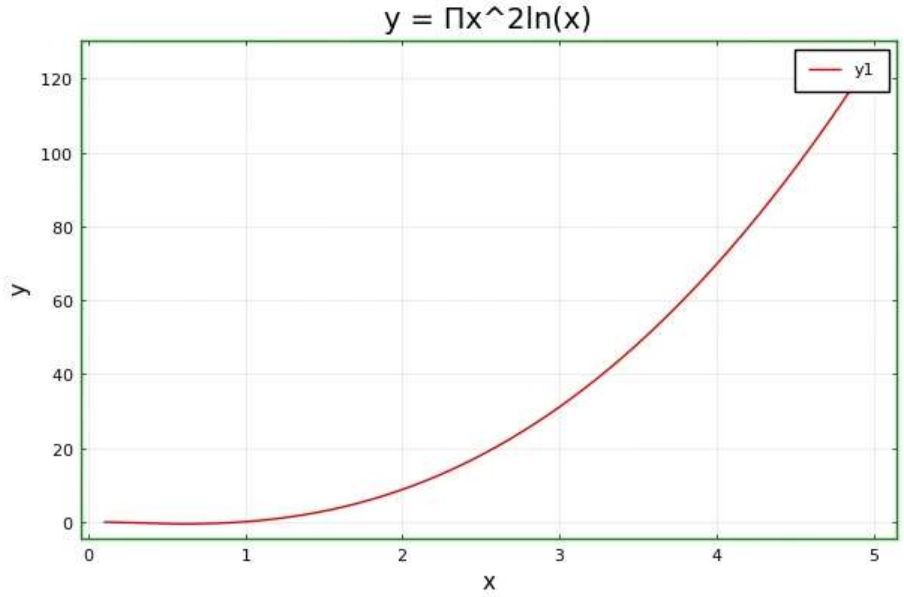


График функции $y(x) = \pi x^2 \ln(x)$

[22]:
```
x = 0.1:0.01:5
y = pi * x.^2 .* log.(x)

plot(x, y,
    color=:red,
    xlabel="x", ylabel="y",
    title = "y = Πx^2ln(x)",
    framestyle=:box,
    foreground_color_border=:green)
```

[22]:



Графики функции $y(x) = x^3 - 3x$

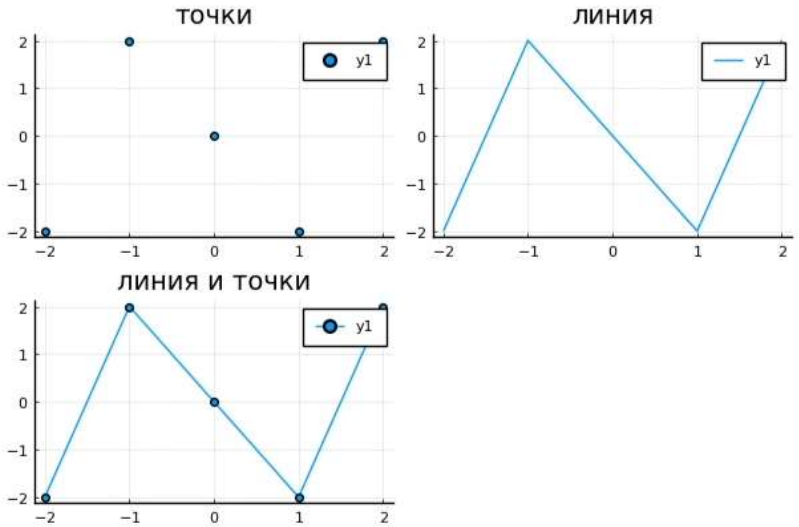
```
•[30]: x=[-2, -1, 0, 1, 2]
       y = x.^3 .- (3 .* x)

       p1 = scatter(x, y, title="точки")
       p2 = plot(x, y, title="линия")
       p3 = plot(x, y, marker=:circle, title="линия и точки")

       xd = range(-2, 2, length=200)

       Plot = plot(p1, p2, p3)
       savefig(Plot, "figure_sacvelev.png")
```

```
sys:1: UserWarning: No data for colormapping provided via 'c'. Parameters 'vmin', 'vmax' will be ignored
```



Решение СЛАУ

```
[23]: A = [1 1 1; 2 2 -3; 3 1 1]
      b = [2; 4; 1]

      A \ b
```

```
[23]: 3-element Vector{Float64}:
       -0.5
        2.5
        0.0
```

```
[24]: A = [1 1 1; 1 1 2; 2 2 3]
      b = [1; 0; 1]

      A \ b
```

Графики функций y1(x) = πx и y2(x) = exp(x) cos(x)

[34]:
```
x = 3:0.1:6
y1 = pi .* x
y2 = exp.(x) .* cos.(x)

plot(x, y1, label="Πx", color=:red)
plot!(x, y2, label="exp(x)cos(x)", color=:blue)
```

[34]:

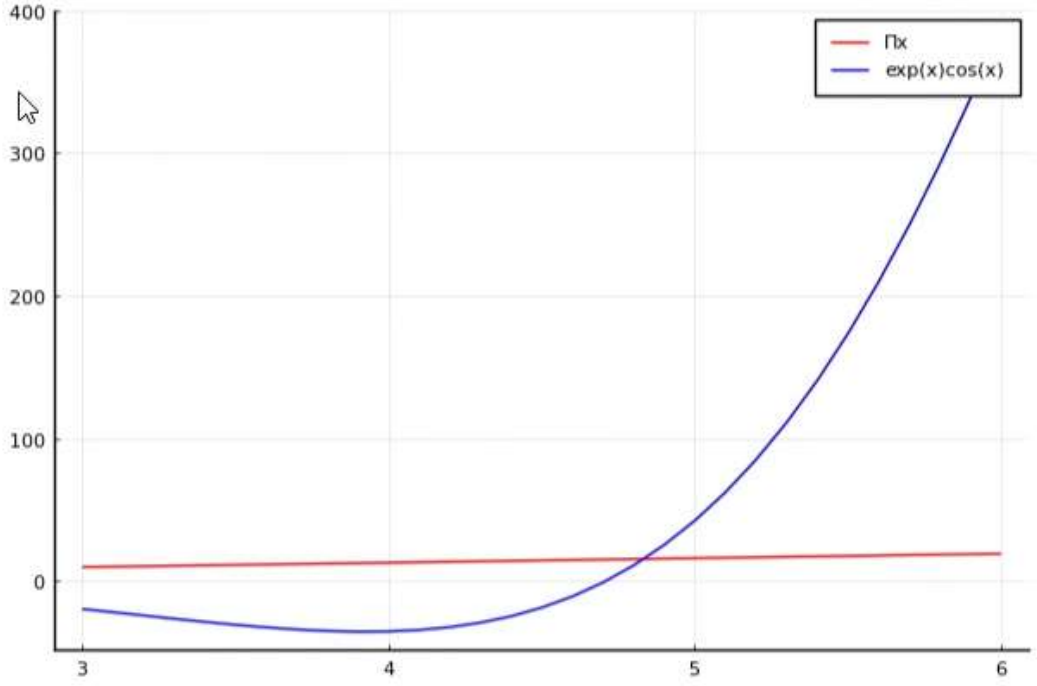
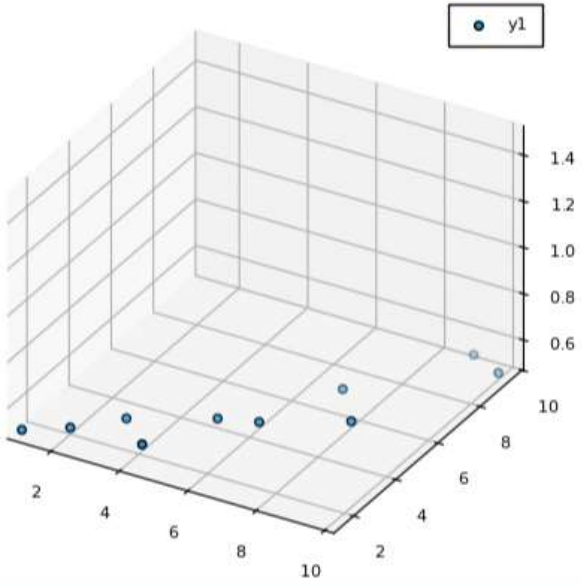


График некоторых экспериментальных данных

[36]:
```
x = 1:10
y = x .+ randn(10)
err = fill(0.5, 10)

scatter(x, y, err)
```

[36]:




```
sys:1: UserWarning: No data for colormapping provided via 'c'. Parameters 'vmin', 'vmax' will be ignored
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sys:1: UserWarning: No data for colormapping provided via 'c'. Parameters 'vmin', 'vmax' will be ignored
```
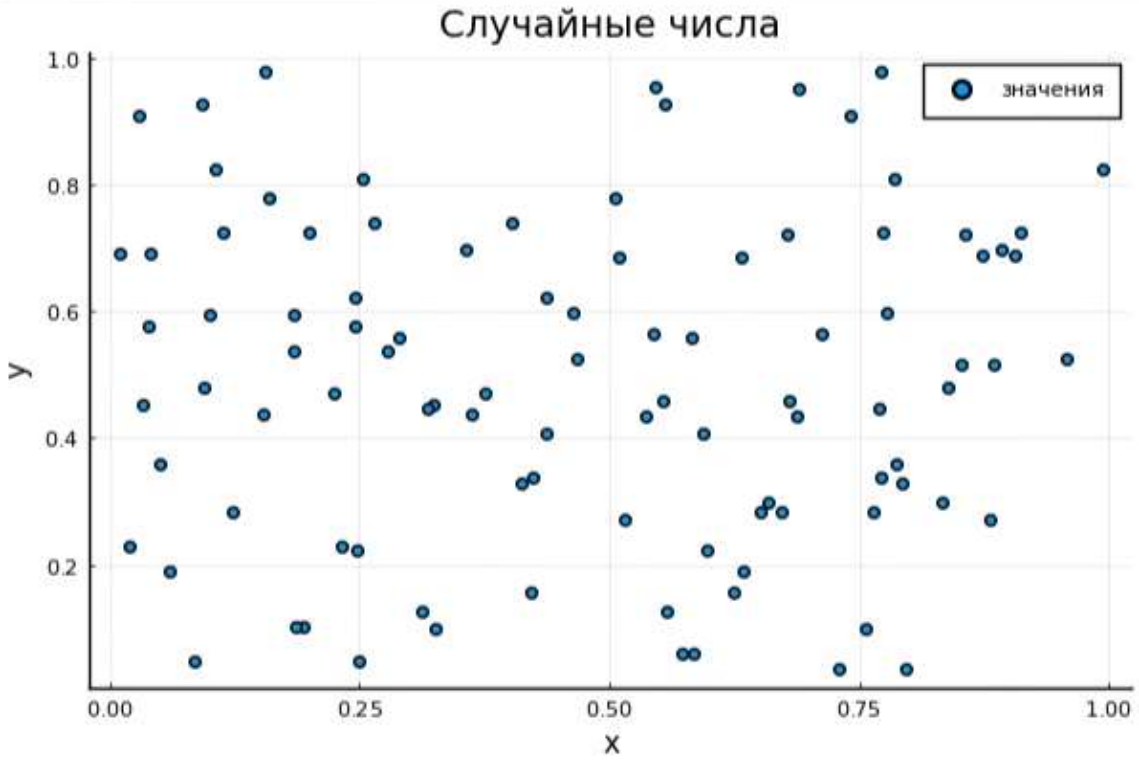

Точечный график случайных данных

```
[37]: x=rand(100)
      y=rand(50)
      scatter(x, y, xlabel="x", ylabel="y", title="Случайные числа", label="значения")
```

[37]:



3-х мёрный точечный график

```
[38]: x=rand(50)
      y=rand(50)
      z= rand(50)
      scatter(x, y, z, xlabel="x", ylabel="y", zlabel="z", title="Случайные числа", label="значения")
```

[38]: